ALGEBRA 1 · STATISTICS AND DATA MODELING

Line of Best Fit Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Line of Best Fit Lab** on Mr. K's Classroom: Algebra 1 → Statistics and Data Modeling → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Hours studied and test scores: positive, negative or no correlation?

2. What do you think happens to r if a student studied 8 hours but scored 40?

2 Explore use the tool and record what you see

Correlation

Record r from the tool.

Situation	r	Strength and direction
Original data		
After adding the outlier		
Points dragged to make r negative		
Points dragged to make r near 0		

Your line vs the best line (original data)

Drag your line, then show the line of best fit.

	Equation	Total squared error
Your line		
Line of best fit		

3 Write the rule fill in each blank from what you observed

1. The correlation coefficient r is always between _____ and _____.
2. r close to 1 means a strong _____ correlation; close to –1 a strong _____ correlation; close to 0 _____.
3. A residual is the _____ distance from a point to the line: actual – predicted.
4. The line of best fit makes the sum of the squared residuals as _____ as possible.
5. An outlier can _____ the value of r and pull the line toward itself.

4 Check no tool: show your work

1. Use y = 4.67x + 49.3 to predict the score for 9 hours of study.

2. What does the slope 4.67 mean in this situation?

3. A student studied 5 hours and scored 68. Find the residual.

4. r = 0.9 for ice cream sales and sunburns. Does ice cream cause sunburns?

5 Reflect

Why do we square the residuals instead of just adding them?